

@josesilesdata



5 Data Engineering Project Ideas

Completely **FREE**

SWIPE TO

Become a Real **Data Engineer**



The Automated Batch Pipeline

GOAL

Extract data from public API, clean and transform it, and load it into a local relational database on a scheduled basis.

TECH STACK

- Python
- Docker
- PostgreSQL
- Apache Airflow



LEARNINGS

- Writing robust API extraction scripts.
- Basic data modeling and SQL schema design.
- Orchestrating and scheduling tasks using Airflow.

RESOURCES

- Docker Desktop: To run Airflow locally
- Public APIs OpenWeatherMap



The Modern Data Warehouse

GOAL

Build a local data warehouse by ingesting raw CSV files into a fast analytical database, then use a data transformation tool to build reporting-ready tables.

TECH STACK

- dbt
- Python
- DuckDB
- Metabase



LEARNINGS

- Using dbt for modular data transformations.
- Dimensional modeling (Fact/Dimension tables).
- Connecting Metabase (BI tool) to local warehouse.

RESOURCES

- DuckDB Documentation: The "SQLite for Analytics"
- dbt fundamentals course
- Kaggle Dataset e-commerce



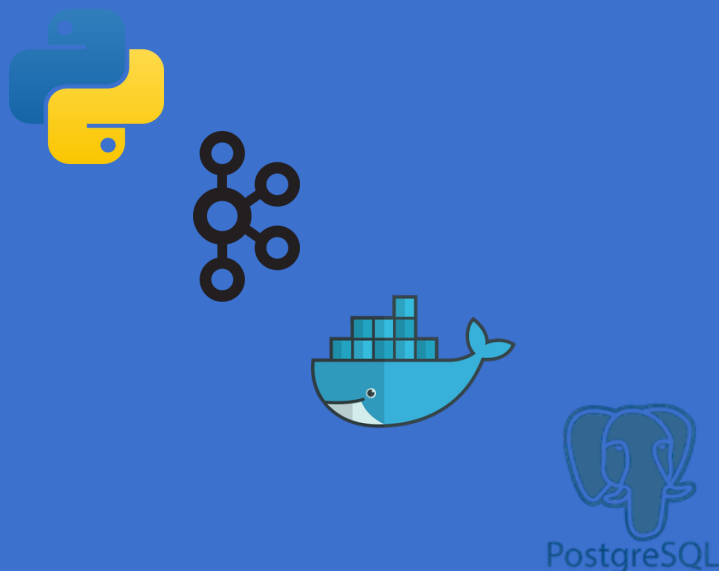
The Real-Time Streaming Engine

GOAL

Simulate a stream of real-time events, capture them in a message broker, and process them as they arrive before saving them to a local database.

TECH STACK

- Python
- Kafka
- Docker
- PostgreSQL



LEARNINGS

- Difference between batch and streaming data.
- Kafka Producers (send) and Consumers (read).
- How Publish/Subscribe messaging systems work.

RESOURCES

- Kafka Quickstart: Run Kafka locally via Docker
- Faker Library: Generate fake stream data



The Local Data Lakehouse

GOAL

Process massive datasets using distributed computing frameworks locally, saving the output as optimized Parquet files in local object storage.

TECH STACK

- MinIO
- Apache Spark
- Apache Iceberg



LEARNINGS

- How object storage works using MinIO.
- How Big Data engines process data in parallel.
- How open table formats Iceberg work on flat files.

RESOURCES

- MinIO Quickstart: Guide to spin up local storage.
- PySpark Documentation: Spark's Python API guide.
- NYC Taxi Dataset: A free massive dataset.



The Web Scraper & NoSQL

GOAL

Build a web scraper to systematically extract unstructured text and image links from a website and load it into a local document database.

TECH STACK

- Python
- Docker
- MongoDB
- BeautifulSoup



BeautifulSoup

LEARNINGS

- Parsing and extracting unstructured web data.
- Handling website pagination and request limits.
- Working with JSON documents in a NoSQL database.

RESOURCES

- BeautifulSoup Docs: Official quickstart guide
- MongoDB Docker: Running MongoDB locally





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